

Course Code: CS2800

Name of the Programme: B. Tech (Hons.) CSE

Name of the Course: Probability and Statistics

Semester: IV

Course credit	No. of hours per week (L + T + P):	Total no. of Teaching hours:
4	3+1+0	45L+15T

Course Objectives:

- To review fundamental concepts of probability theory like probability spaces, sample space, relative frequency, conditional probability, Bayes' Theorem, independence of events.
- Discuss discrete random processes described by discrete random variables and illustrate different pmf like Geometric, Binomial, Multinomial and Poisson distributions and calculate useful quantities like expectations, moments, variance of a sum, correlation coefficient,
- Discuss random processes described by continuous random variables and calculate probabilities and expectations using cdf and pdf.
- Describe regression and CLT to be used in sampling-based hypothesis testing including z-score, critical value, p-value, t-test and for illustrate chi square goodness of fit.

Syllabus:

Module – 1: Probability Basics	No. of hours
	8
Probability spaces, sample space, relative frequency, conditional probability, Bayes' Theorem, independence of events. Simple random walk (one dimensional, two dimensional), distribution of the fortune, Gambler's ruin problem.	

Module – 2: Discrete Distributions	No. of hours
	12
Discrete random variables - Independent random variables; Bernoulli trials, Geometric, Binomial, Multinomial and Poisson distributions; Poisson approximation to the binomial distribution, joint distributions, sums of independent random variables, expectation of discrete random variables, moments, variance of a sum, correlation coefficient, Chebyshev's Inequality.	

Module – 3: Continuous Distributions	No. of hours
	12
Continuous random variables- properties, cumulative and density distribution functions- normal, exponential and gamma densities. Bivariate distributions and their properties; distribution of sums and quotients; conditional densities and expectations (Bayes' rule).	

Module – 4: Regression	No. of hours
	5
Curve fitting by the method of least squares- fitting of straight lines, second degree parabolas; correlation and regression - Rank correlation (Spearman and Pearson).	

Module – 5: Hypothesis Testing	No. of hours
	8
Sampling distribution, Central Limit Theorem, null hypothesis, alternate hypothesis, z-score, critical value, p-value, t-test; Chi-square test for goodness of fit and independence of attributes.	

Course outcomes: After completing the course, students will be able to:	
a)	CO1: Apply discrete probability distribution functions to solve real world problems.
b)	CO2: Apply continuous probability distribution functions to solve real world problems.
c)	CO3: Apply the method of least squares to fit a data set, and interpret the results in the context of the problem at hand.
d)	CO4: Test a given hypothesis using methods like critical value, p-value and chi-square tests for goodness of fit and independence of attributes.

Text Books:	
1	S. Ross, <i>A First Course in Probability</i> , 10th ed. Noida, India: Pearson Education, 2022. ISBN: 978-9356064034.
2	W. Navidi, <i>Statistics for Engineers and Scientists</i> , 4th ed. New York, NY: McGraw-Hill, 2015. ISBN: 978-0073401331.
3	E. Kreyszig, <i>Advanced Engineering Mathematics</i> , 10th ed. Hoboken, NJ: John Wiley & Sons, 2011. ISBN: 978-0-470-45836-5.

Reference books:	
1	W. Feller, <i>An Introduction to Probability Theory and its Applications</i> , Vol. 1, 3rd Ed., Wiley, 1968
2	Shafer and Zhang - Introductory Statistics -LibreText https://stats.libretexts.org/Bookshelves/Introductory_Statistics/Introductory_Statistics_(Shafer_and_Zhang)) , ISBN: 978- 1453344859.
3	N.P. Bali and Manish Goyal, <i>A Textbook of Engineering Mathematics</i> , Laxmi Publications, Reprint, 2010.
4	P. G. Hoel, S. C. Port and C. J. Stone, <i>Introduction to Probability Theory</i> , Universal Book Stall, 2003 (Reprint).

Tutorials/Activities:		[15 Hours]
1.	Tutorial 1: Probability spaces, sample space, relative frequency	
2.	Tutorial 2: Bayes' Theorem	
3.	Tutorial 3: Random walk (1D, 2D), distribution of the fortune, Gambler's ruin problem.	
4.	Tutorial 4: Discrete random variables - Independent random variables;	
5.	Tutorial 5: Discrete Distributions (Bernoulli trials, Geometric, Binomial, Multinomial and Poisson distributions)	
6.	Tutorial 6: Joint distributions, sums of independent random variables,	
7.	Tutorial 7: Expectations, moments, variance of a sum, correlation coefficient, Chebyshev's Inequality	
8.	Tutorial 8: CDF and PDF - normal, exponential and gamma densities.	
9.	Tutorial 9: Bivariate distributions;	
10.	Tutorial 10: Distributions of sums and quotients;	
11.	Tutorial 11: Conditional densities and expectations (Bayes' rule).	
12.	Activity 1: Curve fitting by the method of least squares- fitting of straight lines using Octave	
13.	Activity 2: Second degree parabolas; correlation and regression - Rank correlation (Spearman and Pearson) using Octave.	
14.	Activity 3: Hypothesis testing using, z-score, critical value, p-value, t-test in Octave	
15.	Activity 4: Chi-square test for goodness of fit and independence of attributes.	